

Customer Segmentation Project Report

Introduction

Customer segmentation is the process of dividing customers into groups based on shared characteristics and purchasing behavior. It helps businesses understand their customers and create targeted marketing strategies.

Objective

To segment customers based on annual income and spending behavior using K-Means Clustering and derive actionable business insights.

Dataset Description

Mall Customers Dataset containing 200 customer records and 5 attributes: CustomerID, Gender, Age, Annual Income (k\$), and Spending Score (1-100).

Methodology

1. Data Loading and Exploration 2. Missing Value Check 3. Exploratory Data Analysis (EDA) 4. Elbow Method to determine optimal clusters 5. K-Means Clustering with K = 5 6. Cluster Visualization and Interpretation

Results

Five customer segments were identified:

- Premium Customers (39) – High income, high spending
- Careful Customers (35) – High income, low spending
- Potential Loyalists (22) – Low income, high spending
- Budget Customers (23) – Low income, low spending
- Average Customers (81) – Moderate income and spending

Business Insights

Premium Customers should receive exclusive offers and loyalty benefits. Careful Customers can be encouraged through personalized promotions. Potential Loyalists should be retained through rewards and engagement programs. Average Customers provide upselling opportunities. Budget Customers should be targeted using cost-effective campaigns.

Conclusion

K-Means clustering successfully segmented customers into five meaningful groups. The analysis provides valuable insights for targeted marketing, customer retention, and revenue growth.